

Trileon Triage

From Research Theory to Production Reality

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The Challenge: The "Invisible Child"

As detailed in our *Fusion Framework Document*, standard computer vision algorithms fail in culturally specific healthcare environments.

- **The Occlusion Problem:** Infants carried on mothers' backs cause semantic features to merge.
- **The NMS Failure:** Non-Maximum Suppression (NMS) in standard bounding boxes deletes the child's signature.
- **The Goal:** Build a robust, mathematically explainable pipeline that can differentiate demographics safely and securely.

The Journey: Building the Foundation (MVP)

Before implementing the full PAFs (Part Affinity Fields) and Kalman filters proposed in the Fusion Framework, we required a stable, privacy-first foundational engine.

- **Detection Engine:** We utilized YOLO's native tracking capabilities to establish bounding box kinematics.
- **Middlewares:** Instead of "black box" age estimation, we built an explainable pipeline using Classical CV (Haar/AdaBoost) and **Allometric Canons**.

The Mathematical Core: Allometric Scaling

Human development follows predictable allometric growth. We implemented a linear interpolation algorithm mapping real-time pixel data to biological anchors.

Let R be the measured $\frac{\text{Head Height}}{\text{Total Height}}$.

For $R > \text{Threshold}$ (Pediatric Classification):

$$\text{Confidence} = 0.50 + \left(\frac{R - \text{Threshold}}{\text{Anchor}_{child} - \text{Threshold}} \right) \times 0.49$$

This ensures our confidence scores are derived from anatomical distances, not arbitrary multipliers.

Architectural Code Choices

- 1. Domain-Driven Design (DDD):** By using Facades and strict domain isolation, the Detection logic is separated from the Math logic. This allows us to seamlessly swap YOLO for PAFs in the future.
- 2. JSON Lines (JSONL) Storage:** To prevent memory overflow during 24/7 operation, we use $O(1)$ JSONL logging. Data is flushed line-by-line, ensuring crash resilience.
- 3. Strict Privacy Enforcer:** A recursive algorithm actively blocks image tensors from reaching the disk, guaranteeing patient privacy.

What Didn't Work (And How We Fixed It)

- **Failure 1: Linear Confidence Multipliers.** Initially, we used arbitrary numbers to scale confidence. *Fix: Replaced with the rigorous interpolation mapping to biological anchors.*
- **Failure 2: Video Buffer "Ghosting".** The CPU could not process 30 FPS, causing a massive buffer lag. *Fix: Switched to `cap.grab()` and `cap.retrieve()` to drop old frames at the OS level, keeping the system in true real-time.*
- **Failure 3: Concurrent Counting.** We were counting a child 30 times a second. *Fix: Activated YOLO's internal tracker (`persist=True`) to assign unique IDs to trajectories.*

What Worked Beautifully

- **Stateless Visualization:** The ability to project the bounding boxes and HUD onto the live feed via `cv2.imshow` without ever writing an `.mp4` to disk.
- **The Middleware Pipeline:** The sequence of Haar cascading (fast failure) followed by Allometric Math (geometric truth) proved highly efficient.
- **CPU Throughput:** Frame skipping allows standard hospital desktops to run deep learning models without thermal throttling.

Conclusion & Next Steps

Current State: We possess a production-ready, highly decoupled system capable of accurate Adult vs. Pediatric triage counting under standard visibility.

The Future (Bridging to the Fusion Framework):

- Replace the YOLO base layer with **Part Affinity Fields (PAFs)** to break apart merged pixels during severe maternal occlusion.
- Implement **Inverse Perspective Mapping (IPM)** to dynamically adjust the Allometric threshold as patients walk further from the camera.